Student transfer sheet
Estimator · estimate · sampling distribution

ORIGINAL TRANSFER · STUDENT SHEET

Can you use the idea in a new model?

New fictional model: each independent draw returns 1 point with probability 0.75 or 5 points with probability 0.25. Sampling is with replacement. We use n=4 draws to estimate the fixed population mean μ. A realised sample is **[1,1,5,1]**. These questions are original teaching material, not copied exam questions. Use extra paper for optional calculations.

1. Find μ. Identify a parameter, an estimator before drawing and its observed estimate for this realised sample. What does the next dot in the mean-distribution strip represent?

2. We repeat this four-draw experiment 1000 rather than 100 times. Which changes: the theoretical SE of one future sample mean, the Monte Carlo depiction, both or neither? Give a reason.

3. Keep the rule $\bar{X}$, but change n=4 to n=16. Describe the population mean, $E(\bar{X})$ and SE. Does the next estimate have to equal μ?

4. Compare rules $T_1=\bar{X}$, $T_2=X_1$ and $T_3=\bar{X}+1$ as estimators of μ. Which are unbiased? Does using all four observations automatically remove the offset in T_3? Optional: calculate each theoretical SE.

5. For n=2, list the four ordered samples. Give their probabilities and show that the probability-weighted expected sample mean equals μ. Explain why four equally weighted cases would be wrong.

6. Follow-on CI question: after an interval is computed, does “95% confidence” assign 95% probability to fixed μ lying in that realised interval? Must exactly 95 out of 100 repeated intervals cover? Explain both claims.